

ActInf Textbook Group ~ Cohort 1 ~ Meeting 13

(Feedback, Next steps, Projects)

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hello everyone it is meeting of the active inference textbook group cohort one it is the final meeting of this section it's july 28 2022 and we are going to be uh especially thinking about feedback and structuring on this textbook group on how the first half of the first cohort went and also the ways that over quite literally the coming minutes and hours and days we will be able to implement structures to improve our experience for those of us who are continuing on to part two covert one as well as those who are joining for a new cohort in the onboarding page there are two columns with check boxes and clicking these check boxes will sign you up to continue on in chapter six or ten for cohort one as well as or alternatively to join in chapters one through five with cohort two so select one or both of these boxes if you'd like to be continuing on um that's the onboarding note and again other than that which i hope those who are here or watching live update according to how they want other than that we're going to be talking about future textbook groups and other things like that and then also just on the informational briefly on the future textbook group page here's the link to share with people to join future textbook cohorts so to join in and then also please fill out this feedback form it's really important that even if people think they have limited or non-valuable feedback that they provide it because whether it's on this page directly where everyone can edit and so every thought that somebody has like even if right now they can just edit this page in this more anonymous and asynchronous mode or through direct communication but we're experiencing it and so we have the best insight at this point into how to improve it and the affordances that we're seeing today or the the state updates or however you want to use ontology terms to describe it but what we're seeing today we can do so we should have a fun discussion today on some of those topics and then also look towards project ideas many of which are related to the educational process others which are building on education into research and application and so there will be spaces for those as well let's begin though on this future textbook groups page anywhere where people want to jump in writing or speaking how are we going to summarize

and distill the experience of the last few months in this textbook group and bring it towards improvements in our coming cohorts yeah i think having worked examples um as i've talked about and um discussed with brock like in jupiter notebooks or like even um like simple like did you understand the material like question and answer like quizzes at the end of each chapter or um something like that i think which just would be useful like to gauge your own understanding like if there is like a definition or something that we can directly refer back to in the text that'd be useful okay worked examples yes um [Music] great and the questions could also include little tests of knowledge not every question needs to be throwing a baseball into the deep unknown it can be literally clarifying for someone who believes that an answer is in the known unknown space i i believe there's an answer for this i just want to clarify for my understanding but also once you've clarified that and especially if you did just clarify it then write that example because many of these are opportunities for worked uh examples and working examples work in certain situations for formalisms for example it might also work for formalisms that aren't mathematical like here's the system this way or this phenomena how do these ontology terms map or if this is the blank then what is the other thing okay so we have okay anyone else just go for it or let's just continue to edit and organize and add notes and just have more because many people are having any thoughts on directions to go and just like what is the space of some improvements that we can explore and then how will we be implementing some of these changes and then in cohort two and someone's like it could be this way and they've thought of something that's new and not on this list we added if it's on this list we say awesome we agree join us and do it anyone else with raised hands or we can look more at a few other sections before going to the projects like why she different regimes of uh attention i think one way that we can structure that is by clarifying what practices were explicating and where those practices can take place so then because the reality is and people who are here or listening and replay can give their input many people may not have known what exactly to do to engage with the material there's a textbook there's some regular meetings there's an information resource and it's like an infinite game to structure beneficial engagement and where the rubber hits the road is like being clear about what practices people can engage in so if we have math like not everything has to be fractaled out into sub-sub-sub-groups just including many of the practices like if you want to do art about active inference here's a practice and then if people are engaging in that practice it does like create kind of like a group or a guild engaging in that performance so however people think like what are useful minutes and seconds that actually help learning active and we'll be able to have a perspective on that

question in years from now that's very different based upon better understandings of competencies and the fundamentals and the approachment to the area however i think the practices table and providing feedback and then sharing and communicating how we're using practices to increase understanding could be very important for those who are learning and learning how to learn jessica and then anyone else uh yes um i think to build up on sort of like blue suggestion i'm thinking that like in someone like all the courses that i've taken um you have like all the participants giving sort of like feedback uh as part of the learning process you give feedback on like the other participants work and so it could be that it's like there's an exercise and everybody has to like fill it up and etc and then um the other participants or like it could be like two or three and so not everybody has to do everything um so it's like yeah two three people you read like it's like jessica's like uh answer and give a feedback and you know so on um because in that way by um having to critique or like analyze other people's work it helps you think through on the material itself and yeah so that's like another way um of also like learning again there's also amazing like yeah you can make you could be probably like making a mistake also not applying it correctly but it's like a tool to help you think through the material by analyzing somebody else's work and while also you know initially also having to do on yourself and then hearing also the feedback that other people are going to be giving you so i was kind of like an idea that came to mind like making reference from like other experiences thank you jessica i think that's a great area to keep in mind and to balance as our build and so on like who's on the supply and who's on the demand side of that so to speak like who is actually offering the feedback and then who's the people who are wanting feedback like some of these projects are things where the person was able to be brought in to another project the like noah's and mike's work bringing them into synchronous and asynchronous resources that are already happening provided them with at least a first step for continuing on their journey others are um they're in different areas and so how to keep it open when people have ideas ranging from educational improvements to robotic deployments what kind of feedback they need and then how to balance like the excitement around applying active inference with what it means to learn like there are biochemistry classes it's not all just biochemistry lab but who's to say that's even the best way or anything like that it's just we want to make sure that we keep it all in mind jessica um yeah and i think this um suggests an idea it's not connected to on someone's project or in a 16 project it's more like an exercise so there's like you know short exercise that it's going to be testing you know variational influence or something like that and and then um yeah like every participant in the class go yeah actually does the exercise

and then it's assigned to other people in the cohort and to analyze that person's submission and so it's like yeah three people in the course like analyze like jessica's you know submission and give feedback and like yeah it's gonna be like for everybody everyone and we'll be getting um three um three feedback on like how they apply the terms or like you know there are other ways that they could be done etcetera and i mean if some of the things are like math related like i will have to see and because then the answer is gonna be more clear it's not gonna be as subjective but it's more like you know there's a short exercise everybody does the exercise and then some people then get to review that um you know that answer of the person and give feedback on that so it's like separate from an actual project okay makes sense it's kind of like a combination of having a good example questions with an organizational structure and then like you said balancing like where there might be a clearer uh measuring stick like was this the right equation or the right calculation or not versus um how did this usage of affordances in a new context make sense to you and um yeah also it's not just about getting feedback on the person's idea it's like the practice of providing constructive ideally written comments is super important like the secret side of the questions is the discourse where the kinds of things that questions and ideas that send people on thought excursions and research rabbit holes we can collaborate on that response together and it doesn't have to be the end-all be-all but sometimes by putting like a seed into the workspace by putting like a prompt or a question ten people could write for five minutes and then there would be 50 minutes of writing on that topic from different perspectives whereas if there's 10 people also there could be five minutes of discussion which might add more seeds onto the table and it might continue to invoke whimsy or it might continue to instruct if there was a broad uncertainty that that five minutes of discourse resolved i think like for the questions and it's like somebody has like a question that they're trying to understand and everybody's giving it's the crowd source um answer and helping you know sense make on the meaning of on how things are with the other exercise i think similar to i think what blue wants to do like with the short um like exercises to test whether people know the concept or not it's just like adding like by having people critique other people's work and having to like critically think through the answer is another level of learning because you it's not just like oh i think this is the answer to this it's like no i actually have to understand one day concept that i'm using to evaluate an answer and then also evaluate the person's answer kind of compare and contrast and that like having to make that exercise of critical thinking i think helps people um learn like in a more in-depth way because you have to think through your own thinking process and like what things mean what somebody else was

saying etc and yeah and i i've seen it apply to like writing i mean i guess it's maybe like easy after like writing courses and things like that um but yeah it's like another way of um yeah like processing through the information and and then also um it also maybe like reduces the burden on maybe like a person having to like go through everybody's answer and making sure like yeah like i am the teacher correcting everybody's answers like it's yeah you're sharing the responsibility around everybody in the group of you know like kind of answering you know like you know evaluating people's like responses to an exercise thank you brock and then anyone else with the race hand uh just to agree with jessica and the last part would i was going to add that but um yeah that just systematically created creating a systematic affordance for people to review or provide some validation that it's correct or on the right path or whatever um will reduce the burden there and teaching is probably the best way to learn it's learning how to learn is it's like difficult to do without um like learning self-learning is about teaching yourself to get better at teaching you get better at teaching yourself thanks um which continue just the circle back that does kind of require some metastructure to like create examples or source them or whatever and then provide them so i'm not sure outside of the project that i put on there and maybe it's kind of related to jakobs as well but outside of those two things i'm not sure we have a clear affordance to do that yet so for coming up with examples of yeah like examples to work or whatever yeah like sort either sourcing them and then putting them in a structure that says oh this is for section the chapter one or whatever 1.1 or whatever this is uh so not questions that are related to specific parts of the textbook with people to work through but what kind of table or structure it could be i guess virtually the same but instead of a question a question somebody is asking it's an example for other people to answer maybe with an that they can see like if they want to reveal it or whatever but these does that make sense yes um that's that's what it could be an example to work through in this course it doesn't mean this exact tape but just make sure that we have written it down and everyone can be contributing to like write it down and what the structure is because it's not like this is a um impossible affordance we can spin up the table right now and template it and just start to design it and over the coming weeks we have time to digest all these different inputs to to make it so you know in a few weeks it will fly by but we might be able to be implementing these things i wanted to ask what practices did people feel like they engaged in fruitfully whether it's on here or not just when you feel like you were participating in the active textbook group what did you feel like you were doing and how did it help equation translation that was like the most helpful thing that i did um and i don't think it's there in the well i guess it's kind

of there it was like to copy the equations down but like putting them into natural language was probably like the thing that i did that was maybe most useful for myself and others yep it was a great practice it leads to and scaffolds so many other why is it this way what does this mean what are the examples it's not like the prerequisite per se it's possible to ask questions without having the natural language description but whether it's a basic or a multi-line equation it helps bring consilience to that discussion because there's a non-zero probability that then a term is going to be used loosely or it's going to be used one way then it's going to be conversationally used slightly differently or people will not agree on which ontology terms are invoked or not which can then have every equation start to spiral off into every possible ontology term and they very well might actually connect that way but being able to see what is variational free energy and if there's some term that is being connected to variational free energy that's not being described in the that's fine like least squared error isn't in the linear regression $y = mx + b$ so there can be formal and informal concepts that are super important for that aren't with a symbol but this was a great way to be clear what the symbols actually are what other practices did people feel like they benefited from what would they recommend for a daily schedule for somebody who wanted to learn a lot i think plus one there on uh blues um like writing the equations which is like maybe a super simple but open-ended almost kind of version of jessica's suggestion um but also in general uh just adding stuff to the coda asking questions if you read and you don't ask a question did you really grapple with it at all i don't know um how do we improve thinking by writing in the actual number of typed words that people add in as a proxy during a discussion is one thing but even just afterwards or on their own time frame how can we have 10 people who do read a chapter just scaling numbers each of them could think of several questions ranging from simple i wasn't sure about to simple again like more like an example one like what are the three kinds of this and it's totally known and the answer is fully listed to researchy questions but how could we have many more inputs in terms of the questions that people are asking do people feel like they wrote questions in their own notes but didn't bring it into shared workspace or do people feel like for themselves or or others that they observed that the questions weren't actually phrased um jessica then anyone else yes i mean i was gonna share more about student practices and but i guess like first of all answer like the last question you i mentioned there were things that i didn't put in the general um textbook thing which were more about parsing information and which was like helping me connect the dots and and synthesize an information which ali actually helped me a lot also with some of that and but those things were sort of like personal for me to like try to

make sense of the material and in terms of the student practices initially i was thinking to do like every day and you know like an hour each for like the reading and maybe like an hour for the math and to kind of get foundational things um but then i found it was like better for me to maybe dedicate like a longer period of time on one day or two days for the reading so i'm supposed to like a little bit of the reading more like chunk more and you know two three hours in one day and then maybe like follow up another day with like a longer period of reading as well instead of daily reading a little bit and i think that's more because i guess for memory i mean they do say to do like repetition for memory but i think like for me it was like well really longer and you know have like the material fresh in your mind as like really as opposed to going every day and and then like for the math because i guess there's less of them you know keeping the material price in your mind like yeah like daily i was trying to do that so then it fell off a little bit on on that part but yeah like kind of at least one every day like but four days a week at least like an hour so like trying to focus on the math but yeah that's kind of how my practice you know yeah develop over time through the course so the textbook is for as the like the research people who want to apply in their research if we remember like from the introduction provide readers with the ability to understand and use active inference in their own research we'll have a better sense after part two with a more applied side who might the next cohort of the be in and communicate to over many iterations and you know can go many ways but when people are thinking like in the next few weeks who blue ro okay can we review which questions and understand because i i don't quite understand how like so if we go to the um like if you want to check out the questions check out like chapter two like and a lot of the questions were so if you scroll down um keep scrolling keep scrolling like um i'm trying to think about which ones that were skipped let's see and also there's no skipping there's what we can address in the one hour or two hours right but i mean like we can raise our attention to all the important questions yeah it was just like um and then i realized like most of my questions were like of a largely technical nature so let me scroll through the coda um and see if i can find them but it was chapter two it was like let's see um like the general thing to me is how um among other things prevent question self-censorship and filtering well that question isn't just like this one like genius meets genius perfectly framed specific answer that will be understood in one sentence but that question is like a broader space of prompts that don't necessarily have any prior on whether it has to do with an application or it could be a a technical question well so it would use reference yeah good it was early it was like in chapter two and chapter three and there were like a lot of questions um i'm trying to like look at the specific i'll have to look

at my own code instead of watching you beautiful people but um it was and like it it just happened early like in those meetings um and and it's not that there wasn't time like that we didn't address the questions in order it was like let's skip this because it's technical let's skip this because it's technical so i just stopped putting my questions in because i feel like they're technical like what does this little squiggly symbol mean and and that's why like for me the translation of equations into um natural language was the most useful thing because i feel like all of my my questions are technical like related to so i stopped entering them into the coda because they were just skipped over okay thanks well you appreciate it again i personally hope that we can have structures that uncouple the perceived importance of real-time speech from the contributions that people make so that we can be really clear about like which questions we're going to be addressing in conversation but and and use conversation for what it can bring many things and then 200 questions in chapter two with many upvoters or with whatever mechanism that brings different questions to the four so there isn't even like the plausibility of being skipped it's just a question of like what will we raise up and focus on individually and as a group from the many many questions of different kinds support a beginner with fundamentals and a pre-beginner with background and somebody who's asking research questions as well so i think one simple thing is uh if if you go back to the questions um table like you have a um interest column here interesting yes even better maybe there is a what we will address in the netcoming live stream or a session kind of table that is taken from the questions table but the questions table has no such projection of what is interesting or not in some especially in some one-dimensional kind of like yes or no thing um it's just a separation of random things prompts questions there is some process to get it to what will what will be addressed in the live stream again there's ways to like try to do things like randomization or trying to like ensure that you know we don't continuously ask focus from one person or things like that the popularity doesn't kind of win the day there's more complex ways to do this that are kind of almost like decision making processes like um ephemeral group processes and things like that but um yeah i think just separating them it would be helpful there at least um another thing might be like zoom you could do breakout rooms maybe that's not the format again just uh that's kind of an ephemeral group process thing like perhaps everyone doesn't want to focus on the same questions should there be temporary temporary split averaging of attention and that for some short period of time recombine don't know what that means for a live stream or recording precisely but yeah that's possible solution yeah thanks gather is awesome for these things but it does require clarity around how it's going to be done like mass

one-on-one breakouts mass smaller great breakouts and rearranging people using their spatial location to express different things and assemble into different ephemeral teams okay let's look at some project ideas okay so if for those who are here or not i hope i'm not misrepresenting anything um mike brought up some token economies that's something that we've been actively going towards in active block prints so i'll just add just a note column in active lock for ins as well as as mentioned earlier um working with noaa on some more some related topics audiobook recording of a textbook just as we go through these projects um i've recorded the first chapter and the preface type materials do people think that an audio recording has utility and if so is anyone interested to pursue this avenue with blue and i because it's very defined and it involves primarily the equations having natural descriptions and then just reading and it can go many other directions too but this is like um kind of a clear accessibility affordance for different participants that might be interesting as well as just get out there on like youtube and other channels as having that artifact blue i do think it's useful obviously if i've elected to i didn't realize i signed up to be the participant lead but i'm so happy to do co-lead colleague um but but i do think it's useful as someone who drives like a considerable amount of time like every if it's not like i'm not a very good auditory learner but it does help to like reinforce the concepts um and i i just wanted to mention even if like you're not comfortable perhaps like doing the math or reading the textbook um daniel thinks he can read it until it's perfect um i would prefer to do some video or audio editing um so that's also an option is to do audio editing like where he's like you know you occasionally like stutter daniel does it very infrequently but but there's the occasional tripping over a word or like um saying a word twice like when you lose your place or something like that so video or audio editing um i do it with windows video editor but like audio editing is also another possible affordance there and it would involve like you know reading along with textbook to make sure that it's accurate that no sentences are skipped and then to just splice out any like mishaps or just ask him to re-record the you know markup blanket for precision just so you can place it over the trip trip up if needed yeah thanks definitely we would do a multi-part i i just did one pass on the first chapter no equations etc draft but okay anyways just so people who are interested here or re-watching can know about this okay resources for conducting research or developing active inference agents ways to facilitate the starting of that work for participants just a seed idea a lot of this is going to be explored in part two because it's a recipe for designing the models and we'll have a lot of infrastructure around that as so i hope whoever wrote this continues in that interest builds that starting point for conducting research or developing agents based around

what they learn and see in also typing just directly in the code is a little better than commas because sometimes it's hard to understand what the comments are like referring to like okay complete mathematical overview of active inference derivations of key results maps of equations all awesome thank you jacob for suggesting this i think the closest related works are in the actin phonology and in knowledge engineering and other areas too and i hope that we're on this path however jock up or anyone else wants to connect on that actin foundations um so here we added this page if um brock or anyone else wants to add any describe how some of these comments can be integrated with short answer is i'm not entirely sure the precise way to integrate it with the textbook group they're just um perhaps this is related to questions making questions removing the stigma or whatever like it's just helpful i think to have some affordance or regime of attention or thing that says hey if you don't understand this thing about active inference that's a foundational thing like okay i get entropy but and i get energy like the basic concepts of them but like when you're subtracting them what the hell are we talking about um and then maybe there's you know some follow-on questions about mathematically what entropy is mathematically what is energy or you know what does shannon introduce specifically what is the difference these sorts of things um a it's that an affordance for that but b it's also just acknowledging like um there's a wealth of knowledge already here it's not a complete mystery what those back background foundational things are it's or what learners might want to get out of some relatively short course to like get up to speed and feel real confident reading the textbook and so there's a kind of top-down affordance there too i think that needs to be made for um people that can say hey there's a bunch of equations over or examples over here that are these are the most helpful ones i've found for what bayesian inference really is um how that really works um i guess it would be those two things but if you look into the project um notes there on that project idea um thing it's there's a little more detail of um yeah just a couple bullet points there of just it's it's really hard for people i think to get into a new thing if there's not a model if you do chemistry without bore's model like good luck if you have to jump especially to something like electron probability clouds or something it's just like why would you do that when you have a perfectly good periodic table and simple discrete you know and it's enough even to do basic physics right like newtonian stuff right and so you need a model like i think a lot of history is helpful there very minimal amount but just a this is where this came from this is why it's relevant um it's really hard for people i think to blindly calculate without that sort simple context just kind of made a very rudimentary still kind of scaffolding it um but all it's already linked in that project row but i'll put it in that sub

page there too um super work in progress yeah i mean the idea is just there's these questions are going to be asked over and over and over again there will be people coming from a more mathematical more physics a more neuroscience a more from different backgrounds and so why pretend like that's not happening or reinvent the wheel every time somebody asks a question like just having a central resource where people can get a lot of value but also maybe ask a question that's a little more they might not want to ask and seem like it's too simple or it's too whatever um the live stream okay thank you any other comments or thoughts on this otherwise we'll just continue on for the final minutes textbook enrichment yes i hope everyone continues to annotate figure out also things on their own and explore and like we're building something really powerful with annotation so i hope that people can um continue to be enriching with the structures that we're building but yes so much of what this group has been about has been about textbook enrichment and improvement robot modeling unless anyone has other comments i think learning from jf staying in part two will help and um yeah jessica and then anyone else sorry i think i didn't raise my mind yes well any general comments that people want to provide on the textbook group i think the live streams are super important for the lab and the whole space for a maybe obvious reason the textbook group is equally and these are like the two most important things that can add value to this space and the lab participants so that's it indeed glad that those who have made it this far see that over the right time scales in the right way we see more and more vistas to increase the accessibility and rigor of active inference so i hope we can hold that frame and focus our attention provide these functionalities because if it's stateable it's doable in a way if no one from the first cohort wants to be an active facilitator or steward for textbook group 2 some of these changes will be integrated the more people who want to step in to steward in our participatory context are going to be leverage points for so many more people who come through so there's like multiple levels of affordances for those who see them especially at the end of this section it's been quite a few months any last comments okay i'm gonna stop the recording in three seconds unless somebody wants to say something